

Olist E-Commerce Analytics

Power BI Data Analysis Project

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1. Project Overview

This project analyses the Brazilian Olist e-commerce dataset using Power BI. The report examines revenue, orders, customers, products, sellers, payments, delivery performance, customer reviews, RFM segments and customer retention. The objective is to identify key business trends, risks and opportunities that support data-driven decisions.

2. Project Objectives

- Analyse overall revenue, orders and customer performance.
- Identify top-performing product categories and sellers.
- Evaluate delivery performance, delays and freight costs.
- Analyse payment methods, instalments and customer reviews.
- Segment customers using RFM analysis.
- Measure customer retention using cohort analysis.
- Present actionable business insights and recommendations.

3. Tools and Technologies

- Power BI Desktop – data modelling, DAX and report development
- Power Query – data profiling, cleaning and transformation
- DAX – KPI calculations, time intelligence, RFM and cohort analysis
- Microsoft Excel – source inventory and data-profiling log
- Microsoft Word – project documentation

4. Dataset

The project uses the Brazilian E-Commerce Public Dataset by Olist from Kaggle. It contains information about orders, customers, products, sellers, payments, reviews, order items and geographical locations.

Source files:

- Customers
- Orders
- Order items
- Products
- Sellers
- Payments
- Reviews
- Geolocation
- Product category translation

5. Data Preparation and Cleaning

- Created the pDataFolder parameter to manage all CSV source paths.
- Organised the nine source queries inside the 01_Staging group.
- Disabled loading for staging queries to avoid unnecessary model data.
- Reviewed column quality, distribution and profile information.
- Corrected data types for dates, numbers and text fields.
- Applied Trim and Clean to relevant text columns.
- Checked null values in order and delivery timestamps.
- Retained valid nulls connected to cancelled or unavailable orders.
- Created date-only columns for modelling and time analysis.
- Used referenced queries to build the final fact and dimension tables.

6. Data Model

A star-schema model was created using fact and dimension tables. The fact tables store transactional data, while the dimension tables provide descriptive fields for filtering and analysis.

Fact tables:

- FactOrders
- FactOrderItems
- FactPayments
- FactReviews

Dimension tables:

- DimCustomer
- DimProduct
- DimSeller
- DimDate

Analytical table:

- CustomerRFM

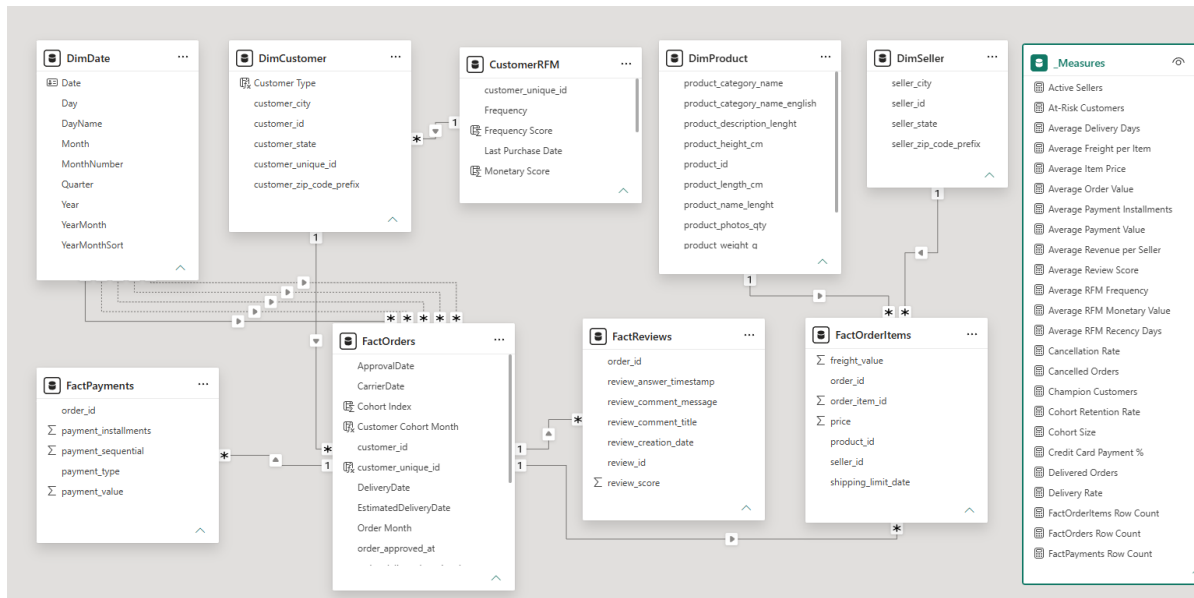


Figure 1: Olist star-schema data model

7. Key Relationships

- DimCustomer → FactOrders
- CustomerRFM → DimCustomer
- FactOrders → FactOrderItems
- FactOrders → FactPayments
- FactOrders → FactReviews
- DimProduct → FactOrderItems
- DimSeller → FactOrderItems
- DimDate → FactOrders

The model mainly uses one-to-many relationships with single-direction filtering. This structure reduces ambiguity and allows dimensions to filter the related transaction tables.

8. Key Measures

Total Revenue – total product price plus freight value

Total Orders – distinct number of orders

Total Items Sold – number of order-item records

Unique Customers – distinct customers

Average Order Value – average revenue generated per order

Repeat Customers – customers who placed more than one order

On-Time Delivery Rate – percentage of orders delivered on or before the estimated date

Average Delivery Days – average time from purchase to customer delivery

Average Review Score – average customer review rating

Positive Review Rate – percentage of positive customer reviews

9. Advanced Analysis

RFM Analysis

Customers were segmented using Recency, Frequency and Monetary value. This analysis identifies loyal, high-value, at-risk and inactive customers for targeted marketing and retention activities.

Cohort Analysis

Customers were grouped by their first purchase month. Cohort index and retention measures were used to track how many customers returned in later months.

Time Intelligence

DAX measures were created to compare revenue across months, quarters and years, including MoM, QoQ and YoY growth.

10. Report Pages

- **Executive Overview** – overall revenue, orders, customers and performance trends
- **Product Analysis** – category and product performance
- **Seller Performance** – seller revenue, orders and delivery results
- **Delivery Analysis** – delivery time, late orders and on-time performance
- **Payment Analysis** – payment methods, values and instalments
- **Customer Analysis** – customer distribution and repeat purchasing
- **Review Analysis** – review scores and customer satisfaction
- **Customer RFM Segmentation** – customer value and engagement segments
- **Customer Cohort Analysis** – customer retention over time
- **Executive Insights** – key findings, risks and recommendations

11. Key Results

- Total revenue: **R\$16.01 million**
- Total orders: **99,441**
- Unique customers: **96,096**
- Total items sold: **112,650**
- On-time delivery rate: **91.89%**
- Average delivery time: **12.6 days**
- Average review score: **4.09**
- Positive review rate: **77.14%**
- Repeat customers: **2,997**
- Repeat customer rate: **3.12%**
- Credit-card share of payment value: **78.34%**
- Month 1 cohort retention: **0.48%**
- Month 3 cohort retention: **0.21%**

12. Business Insights

- Overall sales generated **R\$16.01 million** from **99,441 orders**.
- Delivery performance was strong, with **91.89%** of delivered orders arriving on time.
- Customer satisfaction was generally positive, with an average score of **4.09**.
- Credit cards generated **78.34%** of the total payment value.
- The repeat customer rate was only **3.12%**, showing limited customer loyalty.
- Cohort retention was very low after the first purchase.
- **24,843 customers** were classified as Hibernating.
- **3,784 customers** were classified as At Risk or High-Value At Risk.
- **7,827 orders** were delivered late.

13. Recommendations

- Target Hibernating and At-Risk customers with personalised re-engagement campaigns.
- Introduce loyalty offers to improve the repeat customer rate.
- Investigate sellers, locations and categories connected to late deliveries.
- Monitor freight-heavy products and routes to control logistics costs.
- Improve post-purchase communication and service to maintain positive reviews.
- Continue prioritising credit-card payments while monitoring instalment-related risk.

14. Data Validation

- FactOrders: **99,441 rows**, matching Total Orders
- FactOrderItems: **112,650 rows**, matching Total Items Sold
- FactPayments: **103,886 rows**, matching Payment Transactions
- FactReviews: **99,224 rows**
- Unique reviews: **98,410**
- Duplicate review IDs: **814**
- Order items without a matching order: **0**
- Payments without a matching order: **0**
- Reviews without a matching order: **0**

The validation checks confirmed that the fact tables reconcile with the report measures and that all child records have matching orders. The difference between review rows and unique reviews is caused by duplicate review IDs.

15. Conclusion

This project demonstrates an end-to-end Power BI analytics workflow, including data profiling, cleaning, transformation, star-schema modelling, DAX calculations, validation and dashboard development. The analysis provides clear insights into sales, customers, products, sellers, delivery, payments, reviews and customer retention. The report can support business decisions related to customer loyalty, delivery performance and operational improvement.

16. Dashboard Screenshots

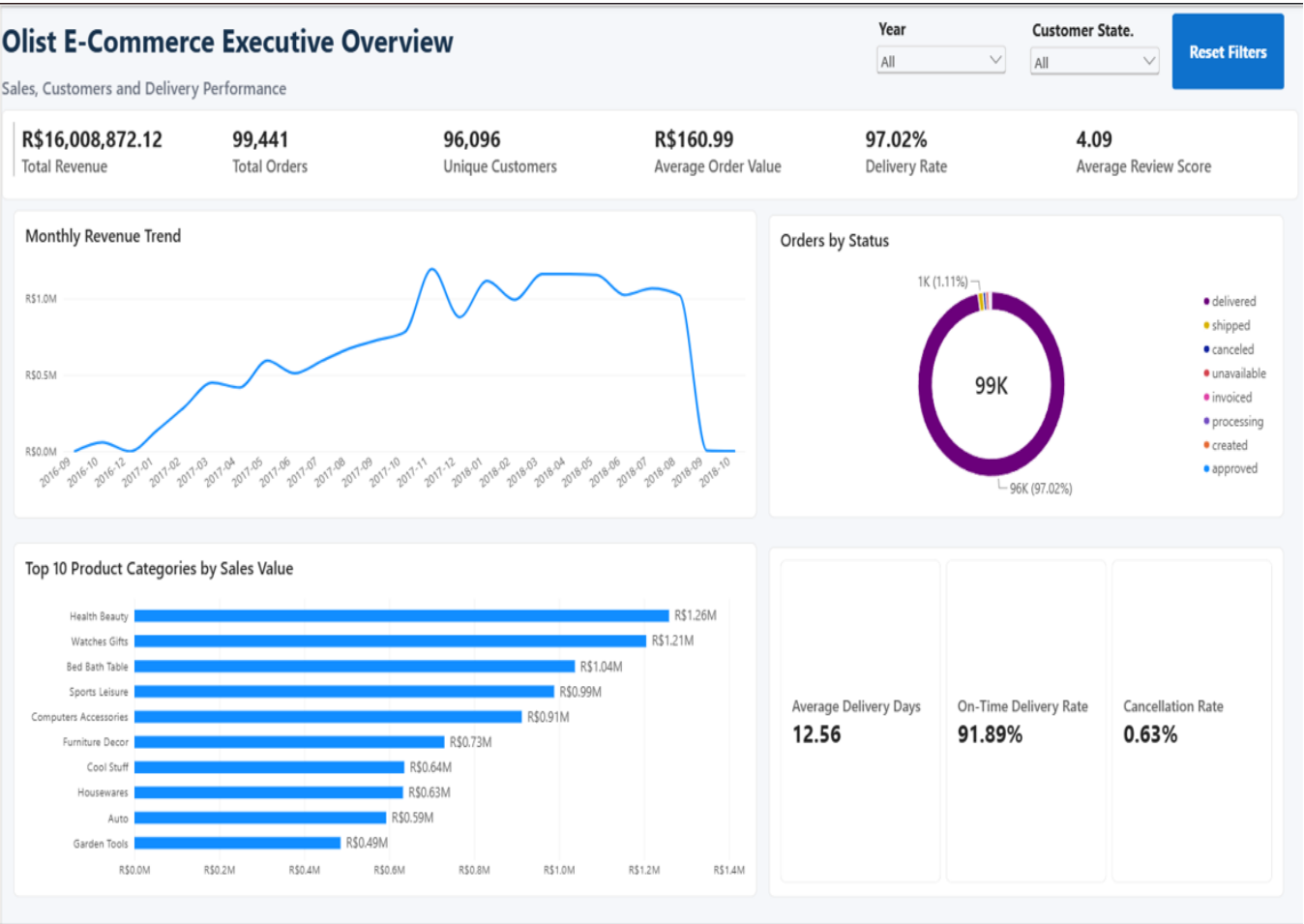


Figure 2: Executive Overview dashboard

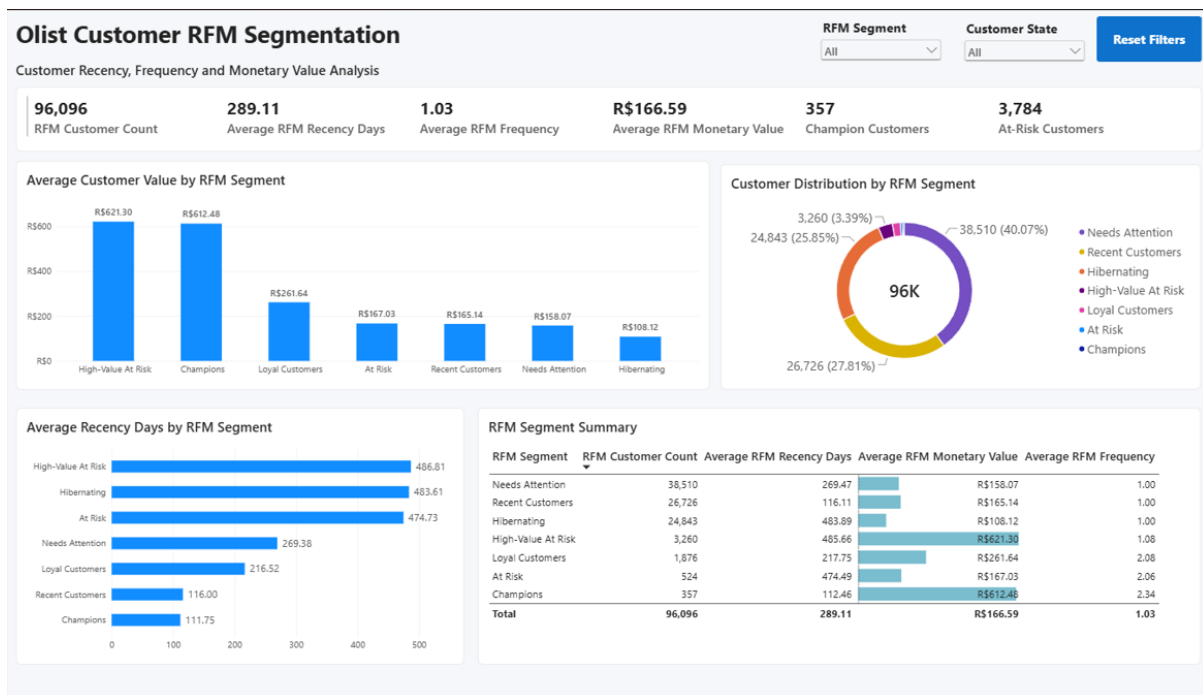


Figure 3: Customer RFM Segmentation dashboard

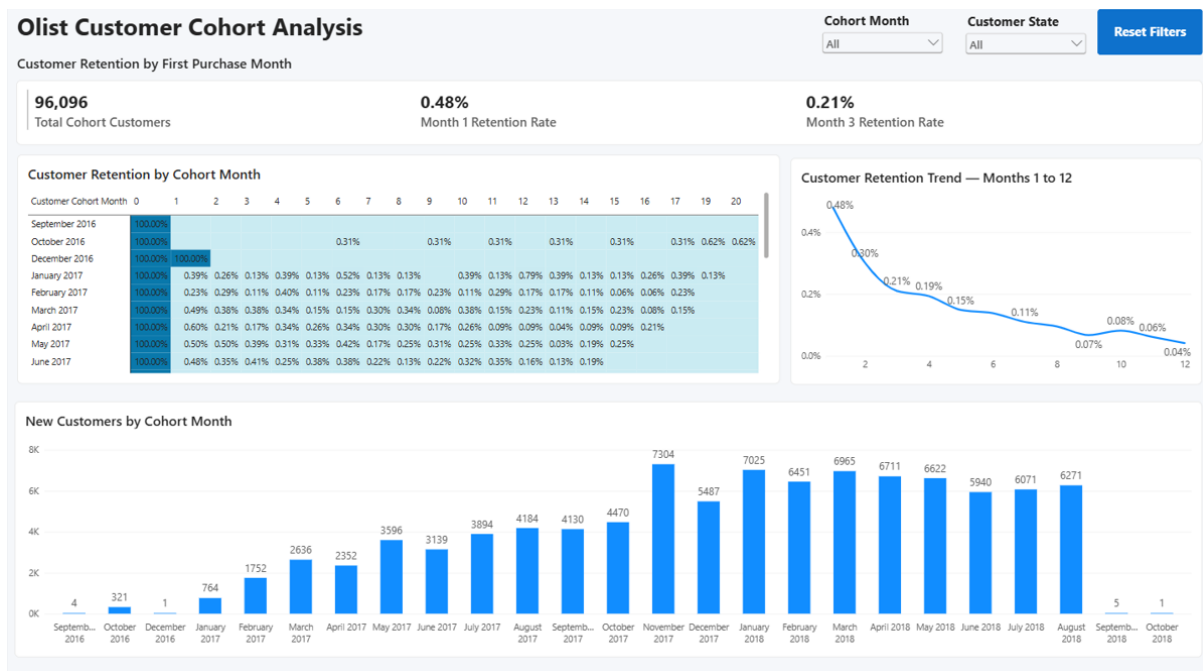


Figure 4: Customer Cohort Analysis dashboard

